



Self-Supervised Pre-training for Skin Lesion Classification under Limited Label Budgets

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Literature Review and Objective

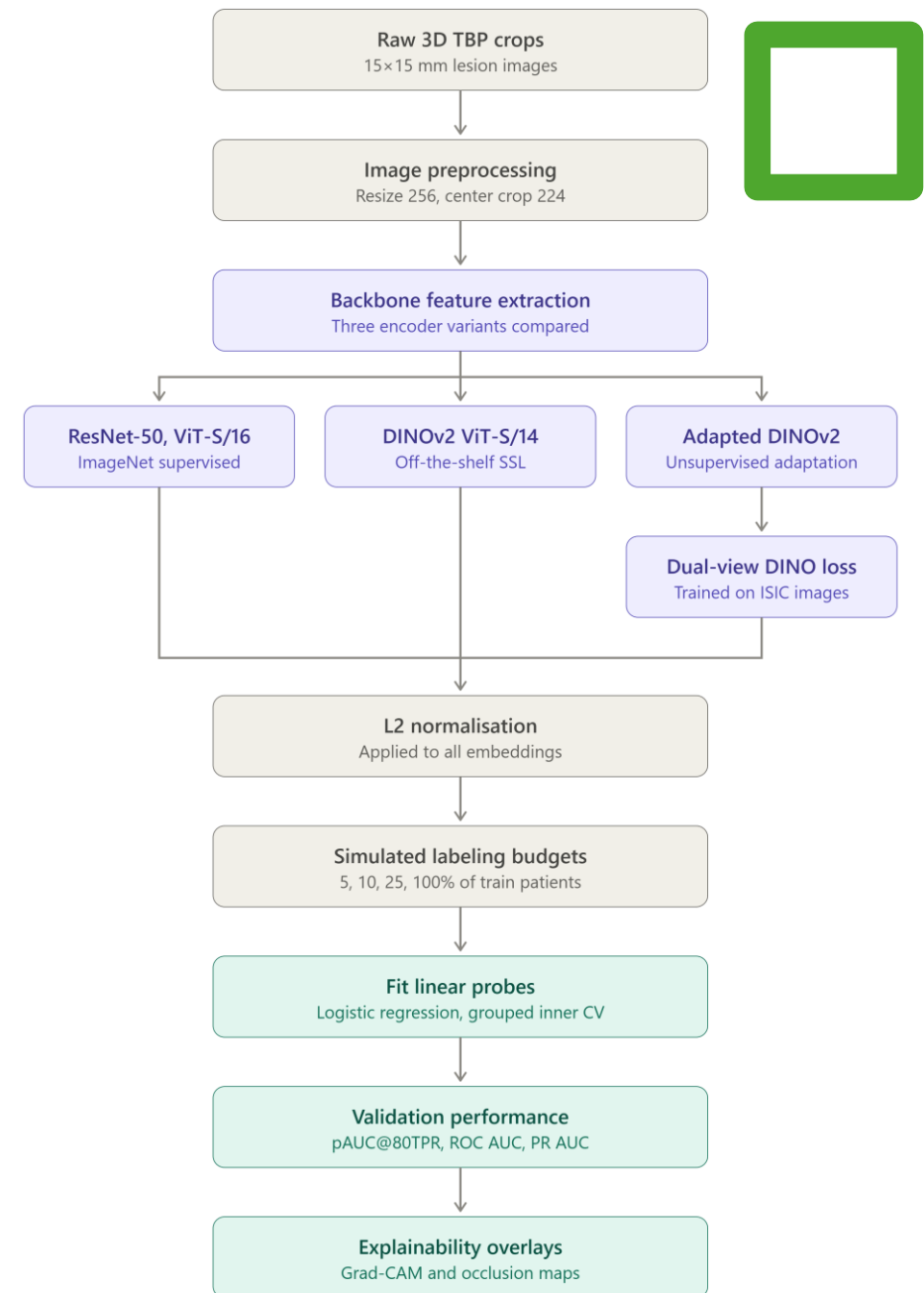
- The status quo at the moment for clinical screening to rely on supervised transfer learning such as vision transformers on generic ImageNet.
- Dermatology needs highly localised, fine-tuned images such melanin distribution and micro textures while the generic ImageNet features are optimised for macro scale natural objects.
- The objective of this research is to evaluate the diagnostic performance and label efficiency of the domain adapted SSL against the traditional supervised learning with expertly annotated data

(Raghu, Maithra, et al)

Dataset Description

- Official Dataset: SLICE-3D Permissive ISIC 2024 (Official Training Dataset of the ISIC 2024 Challenge hosted on Kaggle).
- International Skin Imaging Collaboration (ISIC)
- This dataset has extreme class imbalance:
 - Made up of 217,183 benign cases and only 294 malignant cases (representing an overall cancer prevalence of just $\sim 0.135\%$).
- Training Partition: Consists of a subset of 134,874 images across 381 patients.
- Clinical Realism: The training split contains only 185 positive images, establishing a malignancy frequency of 0.1372% . This highly skewed ratio directly reflects the real-world class imbalance of clinical screenings.

Methodology Flowchart



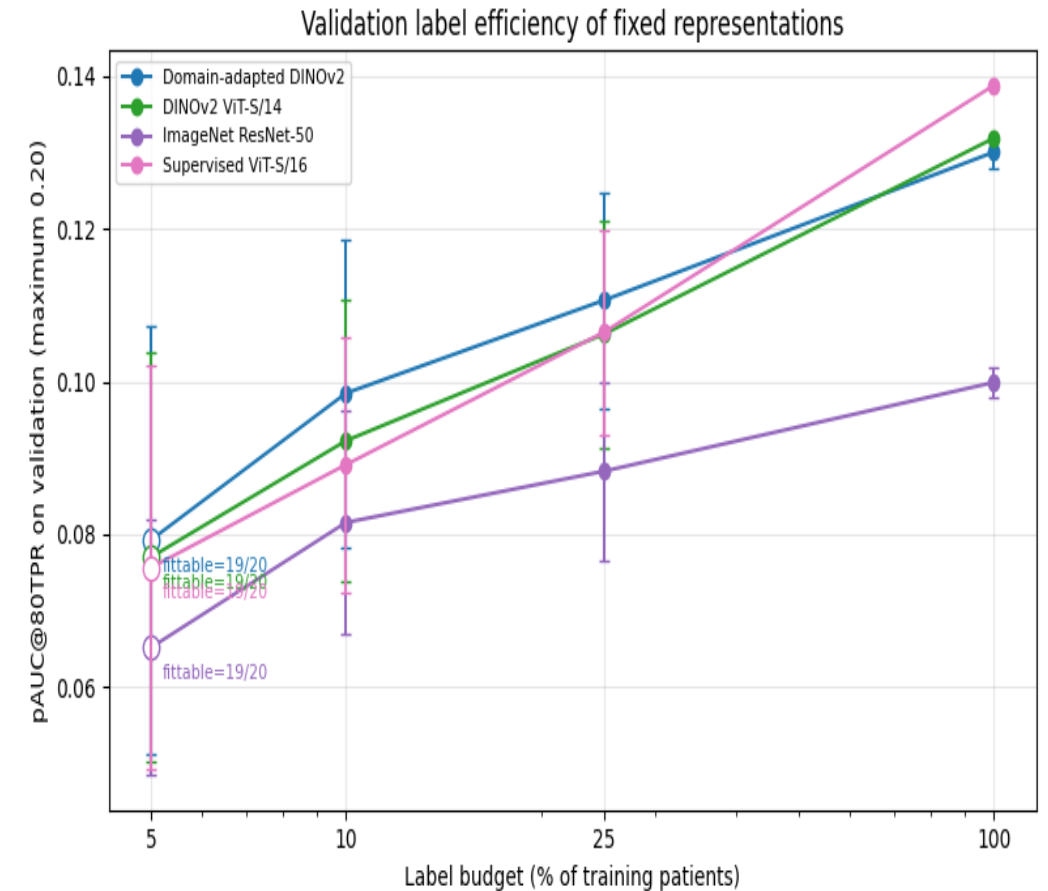
Algorithms and Methodology

Linear probing:

- Instead of training an AI model from scratch on skin lesion images, which is slow, and expensive. This project uses linear probing to analyse frozen visual representation.
- This allows the analysis on feature extractor, if the classifier does well, it proves that the frozen model naturally understand the visual features of the skin lesion with minimal training data.

The four models:

- I used four models to compare which model can create the best representation for the skin lesion
- ImageNet ResNet-50 (The CNN Baseline): A traditional convolutional neural network trained to recognize general, everyday objects (dogs, cars, etc.).
- Supervised ViT-S/16 (The Transformer Baseline): A modern Vision Transformer, also trained on everyday objects.
- Standard DINOv2 : This model is self-supervised learning which is trained on large scale natural image dataset excluding the domain adapted distillation. It learned visual concepts by looking at millions of random internet images without any human-labeled captions.
- Domain-Adapted DINO: The standard DINO model, but specialized. it "look" at unlabelled skin lesion images, so it could adjust to the microscopic details of skin before we started training the classifier.



Means and SDs use all 20 attempted seeds. No-positive budgets receive the constant-predictor chance score. At 100%, seeds vary inner-CV selection only.

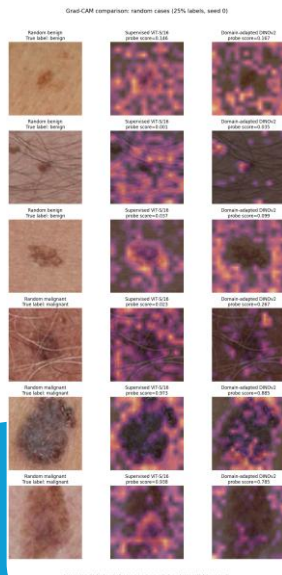
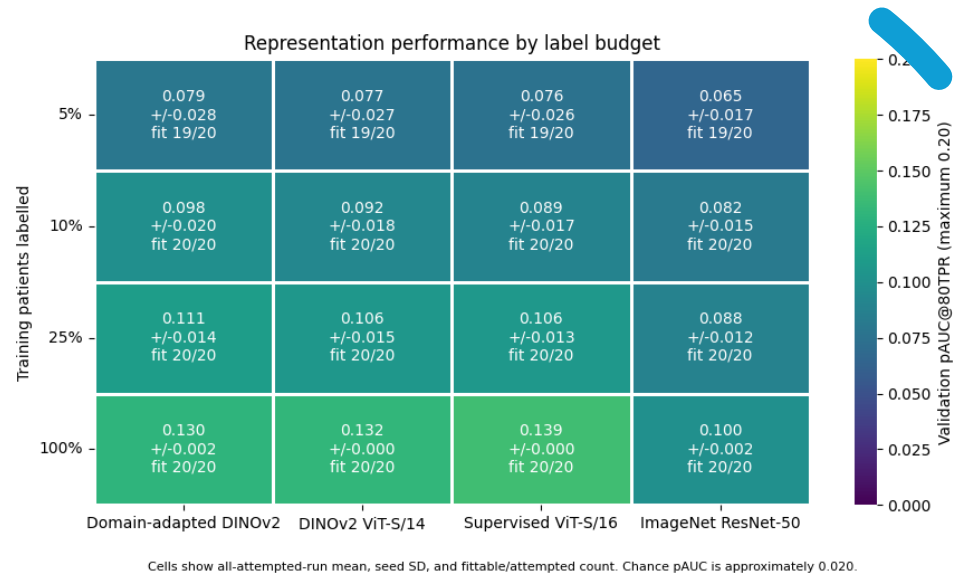
Results

Experimental results:

- 5% Budget: The domain Adapted DINO (0.079) leads off the shelf DINOv2 (0.077), supervised ViT (0,076), and ImageNet ResNet-50 (0.065).
- 10% Budget: The domain adapted DINO (0.098) significantly outperforms supervised ViT (0.089) by statically validated paired bootstrap
- 25% Budget: The domain Adapted DINO leads (0.111) maintains a strong performance lead over ResNet-50 (0.088)
- 100% Budget: Supervised ViT catches up and leads 0.0139 vs the 0.130 for adapted DINO, this highlight the crossover effect which self-supervision becomes the dominate.

Explainability Comparison Grad CAM overlays:

- Both the supervised ViT and the Domain adapted DINO models visuals shows that closely correspond to physical lesion structures. But the domain adapted DINO show a more coherent, spatially localised attention regions instead of fragmented scattering.



Limitations and Future work

Limitations

- Modality dependence: depends a largely on the standardised, 3D total body photography configuration which limits the model's ability to identify the skin lesion.

Future work

- Muti model : ingrate clinical metadata (age, sex, anatomical site and history) directly into self-supervised pre-training to contextually enrich visual representations
- Generalisation Audits: validate diagnostic performance on external, multi ethnic groups to audit clinical safety and ensure algorithmic fairness.